Chapter 1: Causal Inference

1.1 Why is causation important?

We always want to know causal relationships, because we want to know why things around us are the way they are. Why do they keep changing in this way?

In epidemiological, statistical and analytical approaches can broadly be organized into three major domains: descriptive, predictive, and causal analysis.

A) Descriptive: It focuses on describing and quantifying the distribution of exposures, outcomes, and other characteristics within a population.
B) Predictive: Predictive analysis focuses on estimating the probability or risk of an outcome for an individual or population using available information or predictors. It can be used for both diagnostic prediction and prognostic prediction.
C) Causal: Causal analysis focuses on determining whether and to what extent an exposure, intervention, or treatment causes a change in an outcome.

The central question in the causal context is always counterfactual: What would have happened to the same population or individual under a different exposure or treatment?

Now for this project , the last distinction is important, as a causal model that tries to tell you what would happen if we intervened or changed an exposure.

1.2 DAGS(Directed Acyclic Graps)

We will now refer to DAGS, or causal diagrams which is a graph whose nodes (vertices) are random variables V = (V1, . . . , VM) with directed edges (arrows) and no directed cycles.
For eg: If A (treatment /exposure variable), Y- outcome, L is the other variable, the following graph gives us the informations such as

$Y \leftarrow L \rightarrow A \rightarrow Y$,
Where L is a variable which has effect on A and Y .

Another example:

$A \rightarrow L \leftarrow Y$. Here both the variables A and Y lead to L.

Eg:3
$A \rightarrow L \rightarrow Y$, Here the information flows from A to L, then to Y.

These are the three different conditions we get in the causal diagram, where in the first scenario, the variable L is a confounder, where the effect of A on Y is confounded by the variable L.

In the second scenario, L is a collider, as the information from A and Y flows to L, or in other terms L is a future outcome of both A and L.

In the third scenario, L is an intermediate, where the effect on A to Y , is flowing through L.

Why is this important?

In order to know the effect of A on Y, we need to know whether we have any open backdoor paths(confounding paths), we need to adjust for, or are we adjusting for some information in the future(creating biased effect measure), or we adjust for a variable, which creates a spurious relationship.

Components of a DAG

DAGs consist of two fundamental components:

A) Nodes

Nodes represent variables in the causal system. These may include:

- exposures or treatments
- outcomes
- patient characteristics
- biological markers
- behavioral factors
- healthcare-related variables
- selection variables

For example, in a clinical study:

- (E) = exposure or treatment
- (D) = disease outcome
- (A) = age
- (S) = disease severity

may each be represented as nodes in the DAG.

B) Edges

Edges are arrows connecting nodes and represent assumed causal relationships.

For example:

Age → Treatment

indicates that age influences treatment assignment.

Similarly:

Treatment → Infection

indicates the hypothesized causal effect of treatment on infection risk.

The presence of an arrow represents a causal assumption rather than merely an observed association.

1.3 What Information Does a DAG Represent?

A DAG represents an investigator's assumptions about the processes that generate the observed data. These assumptions may include:

1. Determinants of treatment or exposure assignment
   Example:
   Which patient characteristics influence who receives a treatment?
   Disease severity → Treatment
2. Mechanisms through which treatment affects outcomes
   Example:
   Treatment → Biomarker →Outcome
   where the biomarker represents an intermediate pathway through which treatment influences the outcome.

3. Factors influencing the outcome
   Example:
   Age → Outcome

4. Selection mechanisms affecting study inclusion
   Example:
   Disease severity → Hospital admission
   where hospital admission determines whether an individual appears in the observed dataset.
5. Measurement processesDAGs can also represent differences between the true underlying variables and the measurements available to researchers.
   For example:
   True disease status → measured disease status
   recognizing that measurement error may influence causal conclusions.

1.4 Constructing a DAG

Creating a DAG requires careful specification of the scientific question and the assumed causal structure.

The first step is defining the causal question of interest.

For example:

Does treatment (E) reduce disease outcome (D)?

The DAG should include:

- the exposure or treatment ((E))
- the outcome ((D))
- variables that may influence treatment and outcome
- variables that may lie on causal pathways
- variables affecting selection into the study population
- hypothesized causal relationships among these variables

The process of constructing a DAG requires researchers to clearly state their assumptions about the biological, clinical, and behavioral mechanisms underlying the observed data.

1.5 Why DAGs Are Important in Causal Inference

DAGs are not simply illustrations of relationships between variables. They provide a formal framework for answering important causal questions:

- Which variables should be adjusted for?
- Which variables should not be adjusted for?
- Which pathways create confounding?
- Could selection bias occur?
- Are observed associations likely to represent causal effects?

By identifying causal pathways and sources of bias, DAGs guide the selection of appropriate statistical methods, including regression adjustment, propensity score methods, inverse probability weighting, and doubly robust estimators such as AIPTW.

Final Note: DAGs tell us which variables we need to control for; causal estimation methods tell us how to estimate the effect after controlling for them.

Reference

1. Digitale JC, Martin JN, Glymour MM. Tutorial on directed acyclic graphs. Journal of clinical epidemiology. 2022 Feb 1;142:264-7.
2. Tennant PW, Murray EJ, Arnold KF, Berrie L, Fox MP, Gadd SC, Harrison WJ, Keeble C, Ranker LR, Textor J, Tomova GD. Use of directed acyclic graphs (DAGs) to identify confounders in applied health research: review and recommendations. International journal of epidemiology. 2021 Apr 1;50(2):620-32.
3. Greenland S, Pearl J, Robins JM. Causal diagrams for epidemiologic research. Epidemiology. 1999 Jan;10(1):37-48.
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1.6 Causal Inference Foundations – Notes

Motivation : Prediction asks what will happen; causal inference asks what would have happened under a different treatment (counterfactual).

2. Key Variables

A) Unit: patient. Treatment (W): intervention. Outcome (Y): endpoint. Covariates (X): pre-treatment Characteristics.
B) Potential Outcomes : Each individual has Y(1) and Y(0), but only one is observed. The missing outcome is the counterfactual.

4. Treatment Effects

A) ITE: individual effect.
B) ATE: average population effect.
C) ATT: effect among treated.
D) CATE: subgroup-specific effect.

5. Core Assumptions

A) SUTVA (no interference, one version of treatment),
B) Ignorability (no unmeasured confounding after conditioning on X),
C) Positivity (every patient has a non-zero probability of receiving each treatment).

6. Confounding

Confounders affect both treatment and outcome, producing biased estimates if not adjusted.

7. Simpson's Paradox

Aggregated associations may reverse after stratifying by a confounder.

8. Selection Bias

Treatment groups differ systematically; covariate distributions differ between treated and untreated groups.

9. Common Solutions

Stratification, Matching, Propensity Scores, IPTW, Outcome Regression, BART, Causal Forests, TMLE, AIPW, Double Machine Learning.

10. Machine Learning and Causality

Prediction models optimize prediction; causal ML estimates treatment effects while adjusting for confounding.

11. Connection to Synthetic Data

Evaluate whether synthetic datasets preserve causal estimands (ATE/CATE/ATT), causal pathways, and treatment-effect heterogeneity, not just predictive accuracy.

Reference:

1. Li, Sheng & Chu, Zhixuan (Editors). Machine Learning for Causal Inference. Springer, 2023.